

LR

CR

SR

TR

AutoReview Crew

AI agents that review code together and recruit help when needed

Four agents | Three model families | One Band room | 51-second live run

Band of Agents Hackathon | Track 2: Multi-Agent Software Development | Kgotso Msiza

PROBLEM

Code review puts too many checks on one primary reviewer.

Correctness, security, tests, maintainability, and merge risk often land in the same review window.

01

Bottleneck

PRs wait for the busiest reviewer, not the best specialist.

02

Coverage gap

Logic, security, and tests compete for one reviewer's attention.

03

Weak audit trail

The merge decision often loses who checked what and why.

AutoReview Crew turns review into a coordinated team workflow.

Built around Band-native multi-agent software development.

The workflow combines visible agent coordination, PR review, live recruitment, and partner model usage.

Band-native workflow

4 agents coordinate in one shared room

Track 2

PR review, tests, merge preparation

AI/ML API

Lead, Correctness, Test agents

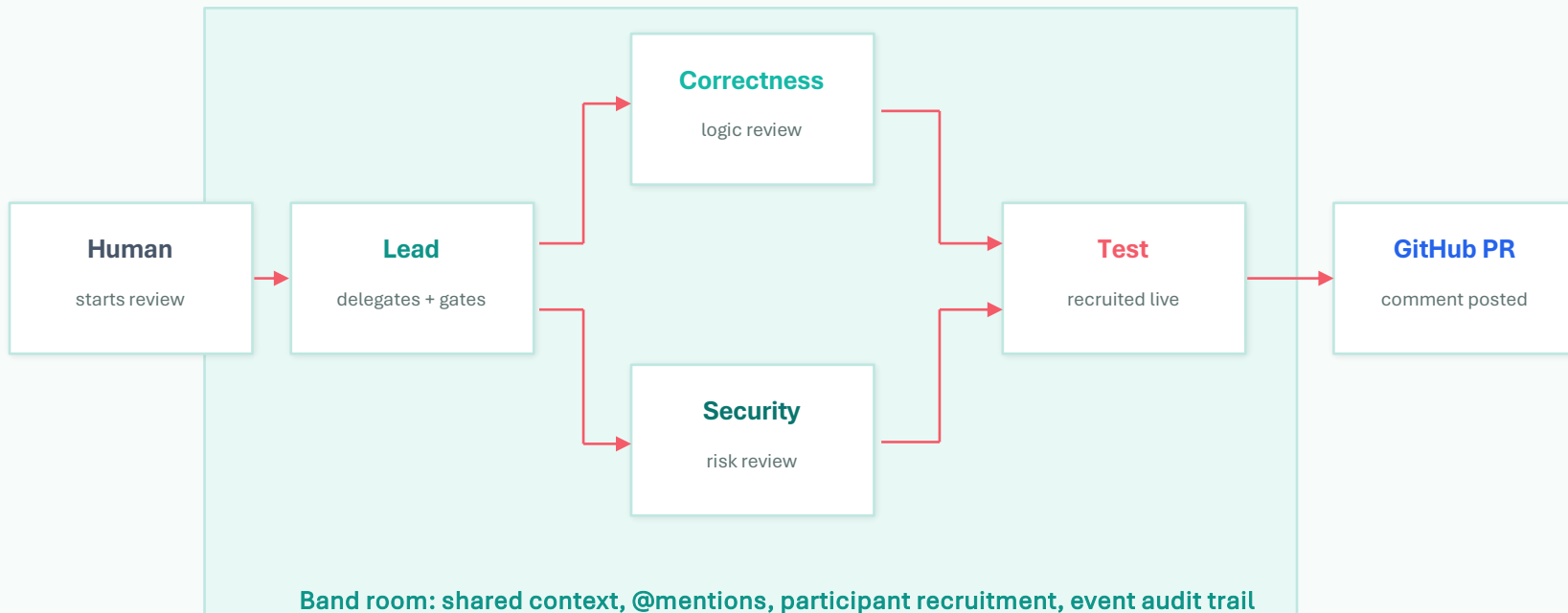
Featherless AI

Security Reviewer on Qwen2.5-72B

Positioning: a Band-native software review room where agents hand off work, recruit specialists, and escalate risky changes with an audit trail.

Band is the coordination layer, not the output channel.

Every assignment, report, recruitment, and final escalation is routed through the shared room.



The submitted video shows the whole workflow in about 51 seconds.

One Lead delegation, specialist reports, a live recruitment, and a final escalation.

13:02:22	Human	@Lead Reviewer please review the code change
13:02:23	Lead	Delegates once to Correctness + Security
13:02:33	Correctness	Finds division by zero + mutable default
13:02:48	Security	Finds hardcoded key, SQL injection, validation gap
13:02:55	Lead	Recruits Test Reviewer through Band
13:03:08	Test	Posts pytest coverage
13:03:13	Lead	Final verdict: ESCALATE_TO_HUMAN

The moment: agents hiring agents

The Test Reviewer is not in the room at the start. The Lead discovers and recruits it only when the gate requires tests.



This is the originality beat: the workflow adapts by adding a new specialist, inside Band, without human intervention.

Different specialists catch different classes of risk.

The Lead aggregates the reports instead of flattening everything into one generic model answer.

Security	CRITICAL	Hardcoded Stripe API key
Security	HIGH	SQL injection
Correctness	HIGH	Division by zero
Correctness	MEDIUM	Mutable default argument
Test	TESTS	Pytest coverage for risky paths

Final decision: ESCALATE_TO_HUMAN because a leaked secret and injection risk should not merge on automation alone.

The build is hardened for live multi-agent messiness.

It is designed to recover from the failure modes we saw while building, not just succeed in one perfect run.

Current-round filter

Old reports from reused rooms cannot satisfy a new review.

All-page context fetch

The Lead sees the whole room context, not just page 1.

One report per trigger

Duplicate tool-call races do not create spam.

Test gate

No final verdict until pytest functions appear.

Started-signal retry logic

Working reviewers get more time; missing reviewers get nudged.

Human escalation rail

Missing specialists or critical findings force human review.

Architecture choice: deterministic coordination; LLMs do specialist review and final wording.

A review crew is faster, safer, and more auditable than a single assistant.

The same Band pattern can expand from PR review into release, compliance, and incident workflows.

For engineering teams

- Specialist coverage on every risky diff
- Human escalation for critical issues
- Audit trail for who checked what
- Faster merge preparation for routine changes

Why it can grow

- Add performance, docs, compliance specialists
- Use org agent registry for reusable expertise
- Post final reports to GitHub or ticket systems
- Apply the same pattern to high-stakes workflows

Four colored circles are arranged horizontally. From left to right: a teal circle with 'LR', a teal circle with 'CR', a teal circle with 'SR', and a red circle with 'TR'.

LR

CR

SR

TR

AutoReview Crew

Specialist AI reviewers that work in parallel, recruit help when needed, and know when to call a human.

Band turns separate agents into a review crew.

Built solo with agentic coding | MIT licensed | Track 2